

Task 1: Configure Git

Verify Git installation and configure your username and email.

Task 2: Create Local Project

Create a new folder for the employee management project and initialize it as a Git repository.

Task 3: Create Employee File

Create a file named employees.txt and add employee details such as ID, Name, and Role.

Task 4: Track Changes

Check the status of your project, add the employee file to tracking, and commit the changes.

Task 5: Create GitHub Repository

Go to GitHub and create a new repository named 'employee-management'.

Task 6: Connect and Push

Connect your local project to the GitHub repository and push your changes.

Task 7: Update File

Add a new employee to the file and update the repository with the changes.

Task 8: Clone Repository

Clone the repository to another location or system and verify the files.

Bonus Challenge:

Create a new branch, make changes, and push it to GitHub.

```
MINGW64/C:/Users/shaur/employee-management
shaur@Shaurya MINGW64 ~
$ git --version
git version 2.52.0.windows.1

shaur@Shaurya MINGW64 ~
$ git config --global user.name "Shaurya Chattopadhyay"

shaur@Shaurya MINGW64 ~
$ git config --global user.email "shauryachattopadhyay80@gmail.com"

shaur@Shaurya MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
core.autocrlf=true
core.fsmonitor=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=Shaurya Chattopadhyay
user.email=shauryachattopadhyay80@gmail.com

shaur@Shaurya MINGW64 ~
$ mkdir employee-management

shaur@Shaurya MINGW64 ~
$ cd employee-management

shaur@Shaurya MINGW64 ~/employee-management
$ git init
Initialized empty Git repository in C:/Users/shaur/employee-management/.git/
```

```
MINGW64~/Users/shaur/employee-management
shaur@Shaurya MINGW64 ~/employee-management (master)
$ cat > employees.txt << EOF
> ID: 101 | Name: Alice Johnson | Role: Developer
> ID: 102 | Name: Bob Smith | Role: Designer
> ID: 103 | Name: Carol White | Role: Manager
> EOF

shaur@Shaurya MINGW64 ~/employee-management (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    employees.txt

nothing added to commit but untracked files present (use "git add" to track)

shaur@Shaurya MINGW64 ~/employee-management (master)
$ git add employee.txt
fatal: pathspec 'employee.txt' did not match any files

shaur@Shaurya MINGW64 ~/employee-management (master)
$ git add employees.txt
warning: in the working copy of 'employees.txt', LF will be replaced by CRLF the next time git touches it

shaur@Shaurya MINGW64 ~/employee-management (master)
$ git commit -m "Add employees.txt with initial employee data"
[master (root-commit) 3fa9f66] Add employees.txt with initial employee data
 1 file changed, 3 insertions(+)
 create mode 100644 employees.txt

shaur@Shaurya MINGW64 ~/employee-management (master)
$ git remote add origin https://github.com/ShauryaChatterjee02/employee-management.git

shaur@Shaurya MINGW64 ~/employee-management (master)
$ git branch -m main

shaur@Shaurya MINGW64 ~/employee-management (master)
$ git branch -m main

shaur@Shaurya MINGW64 ~/employee-management (main)
$ git push -u origin main
remote: Repository not found.
fatal: repository 'https://github.com/ShauryaChatterjee02/employee-management.git/' not found

shaur@Shaurya MINGW64 ~/employee-management (main)
$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 342 bytes | 342.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ShauryaChatterjee02/employee-management.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

shaur@Shaurya MINGW64 ~/employee-management (main)
$ echo "ID: 104 | Name: David Lee | Role: QA Engineer" >> employees.txt

shaur@Shaurya MINGW64 ~/employee-management (main)
$ git add employees.txt
warning: in the working copy of 'employees.txt', LF will be replaced by CRLF the next time git touches it

shaur@Shaurya MINGW64 ~/employee-management (main)
$ git commit -m "Add new employee David Lee"
[main c2d872f] Add new employee David Lee
 1 file changed, 1 insertion(+)

shaur@Shaurya MINGW64 ~/employee-management (main)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
```

```
MINGW64/c/Users/shaur/employee-management
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 346 bytes | 346.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ShauryaChatterjee02/employee-management.git
3fa9f66..c2d872f main -> main

shaur@Shaurya MINGW64 ~/employee-management (main)
$ cd ..

shaur@Shaurya MINGW64 ~
$

shaur@Shaurya MINGW64 ~
$ git clone https://github.com/shauryaChatterjee02/employee-management.git cloned-employee-management
Cloning into 'cloned-employee-management'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 1), reused 6 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
Resolving deltas: 100% (1/1), done.

shaur@Shaurya MINGW64 ~
$ ls cloned-employee-management/
employees.txt

shaur@Shaurya MINGW64 ~
$ cd employee-management

shaur@Shaurya MINGW64 ~/employee-management (main)
$ git checkout -b feature/add-department
Switched to a new branch 'feature/add-department'

shaur@Shaurya MINGW64 ~/employee-management
$ cd employee-management

shaur@Shaurya MINGW64 ~/employee-management (main)
$ git checkout -b feature/add-department
Switched to a new branch 'feature/add-department'

shaur@Shaurya MINGW64 ~/employee-management (feature/add-department)
$ echo "ID: 105 | Name: Eva Green | Role: HR" >> employees.txt

shaur@Shaurya MINGW64 ~/employee-management (feature/add-department)
$ git add employees.txt
warning: in the working copy of 'employees.txt', LF will be replaced by CRLF the next time Git touches it

shaur@Shaurya MINGW64 ~/employee-management (feature/add-department)
$ git commit -m "Add new branch with HR employee"
[feature/add-department 984bf3b] Add new branch with HR employee
1 file changed, 1 insertion(+)

shaur@Shaurya MINGW64 ~/employee-management (feature/add-department)
$ git push -u origin feature/add-department
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 338 bytes | 338.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/add-department' on GitHub by visiting:
remote:   https://github.com/ShauryaChatterjee02/employee-management/pull/new/feature/add-department
remote:
To https://github.com/shauryaChatterjee02/employee-management.git
 * [new branch]   feature/add-department -> feature/add-department
branch 'feature/add-department' set up to track 'origin/feature/add-department'.

shaur@Shaurya MINGW64 ~/employee-management (feature/add-department)
$ |
```